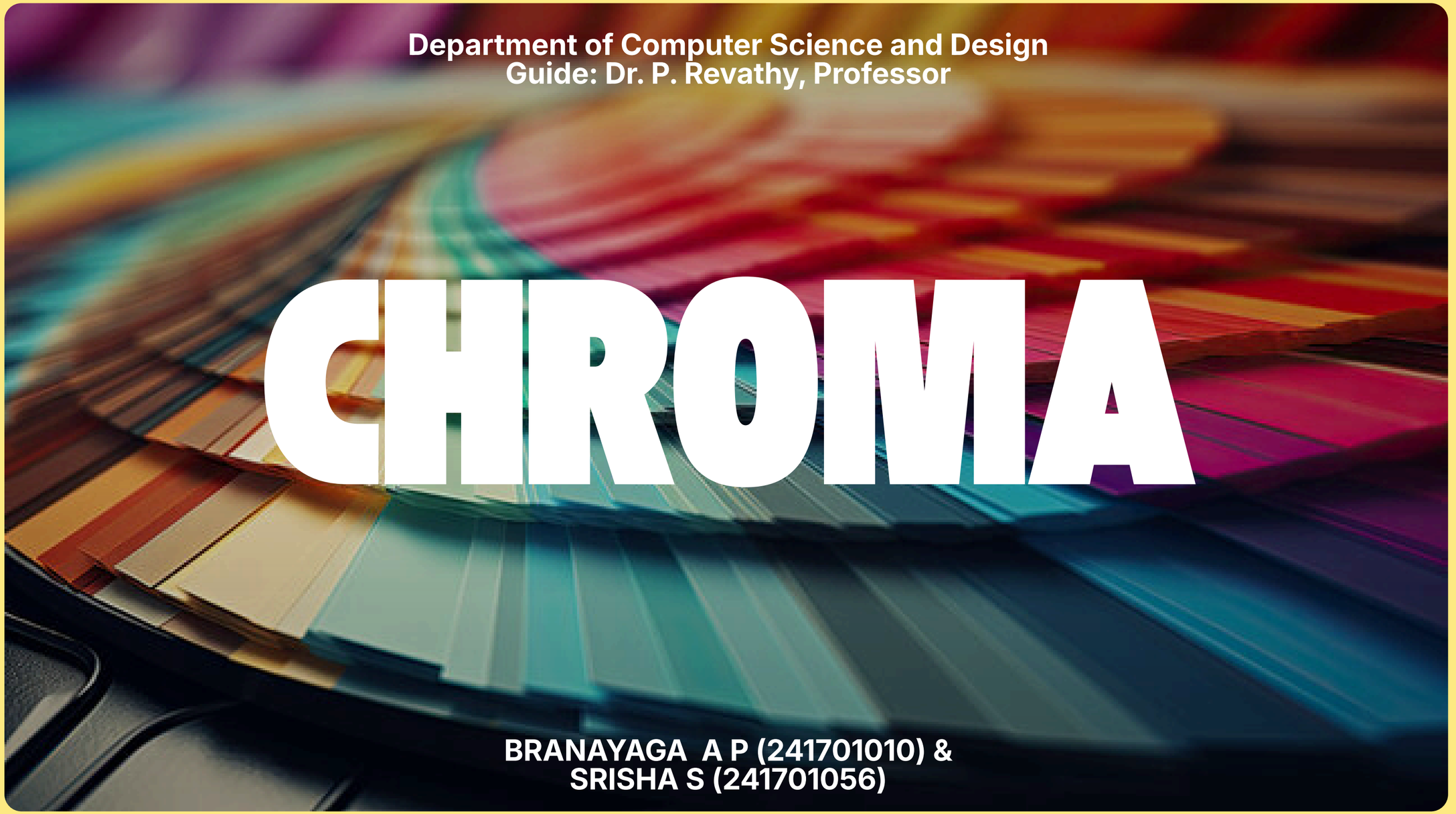
Department of Computer Science and Design
Guide: Dr. P. Revathy, Professor



CHROMA

BRANAYAGA A P (241701010) &
SRISHA S (241701056)

THE PROBLEM



1. Colour Identification

Identifying real-world colours by eye is subjective and affected by lighting conditions.



2. Physical–Digital Gap

Transferring colours from physical objects into digital designs often requires manual colour selection.



3. Colour Matching

Comparing colours across physical surfaces and digital images can be difficult and time-consuming.



4. Colour Visualization

Users cannot easily preview how a selected colour will appear on an actual wall, fabric or object.

OUR SOLUTION

CHROMAPOD
Physical Colour Scanner

A portable device that:

- Captures colours from real-world surfaces.
- Displays RGB/HEX values and colour comparisons.
- Connects wirelessly with CHROMA Studio using BLE.

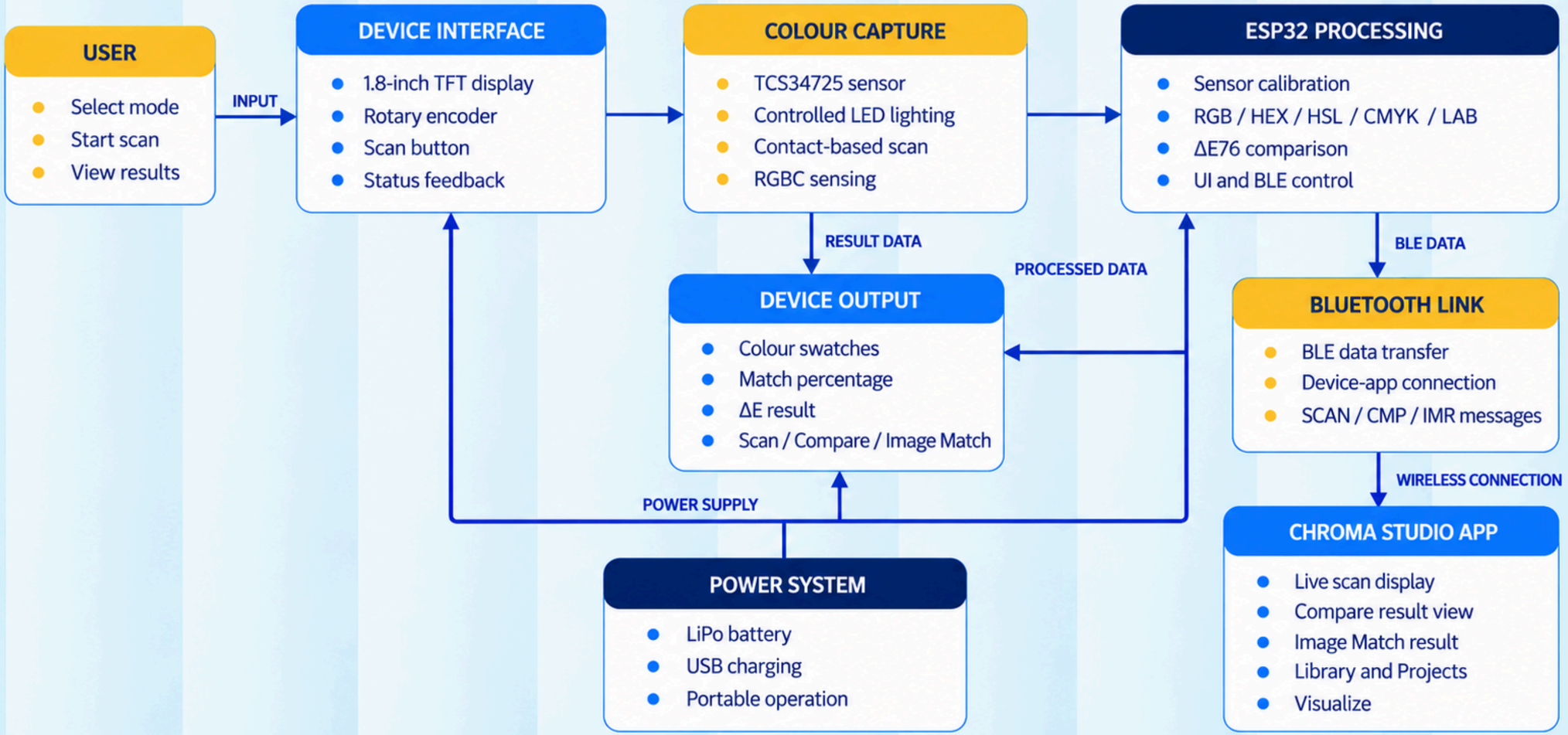
BLE

CHROMA STUDIO
Digital Companion Application

An application that:

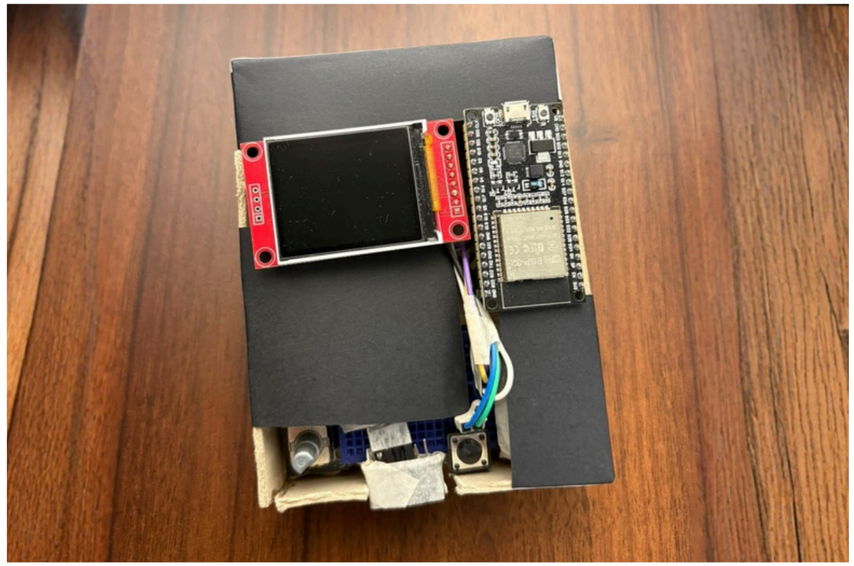
- Analyses, compares and converts colours.
- Performs image matching and surface-colour visualization.
- Saves colours and organises them into projects.

ARCHITECTURE DIAGRAM



MEET CHROMAPOD

THE HARDWARE



ESP32 -Processing & BLE. TCS34725 -Colour sensing. 1.8″ TFT -Display. Encoder + Button -Controls. LED + LiPo -Illumination & power.

THE MODES

SCAN - COMPARE - IMAGE MATCH

01 SCAN

Captures physical colours and displays values.

02 COMPARE

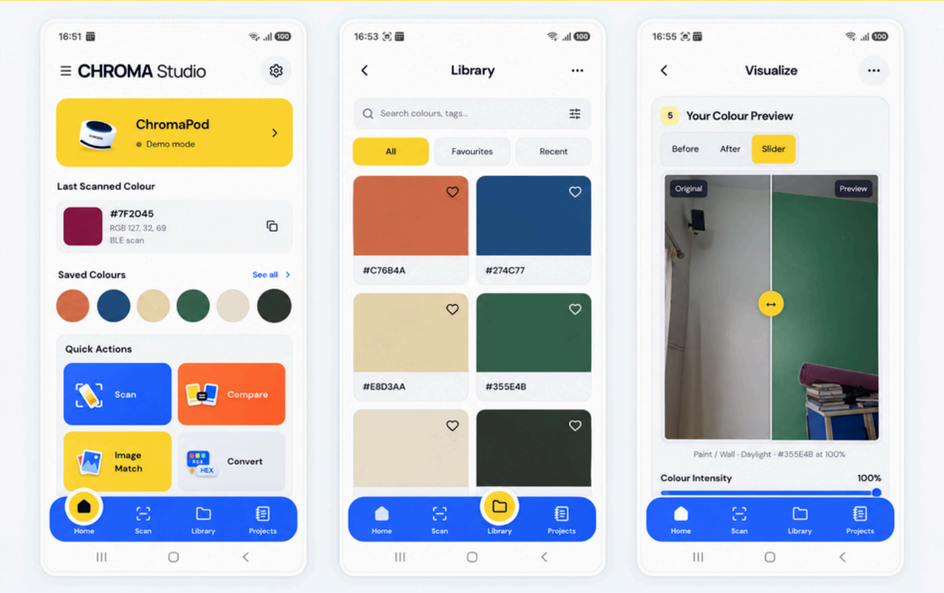
Compares two scanned colours using CIE76 (ΔE76).

03 IMAGE MATCH

Matches scanned colours with image colours via BLE

MEET CHROMA STUDIO

THE APPLICATION



Digital companion application that connects with CHROMAPOD through BLE for colour analysis and management.

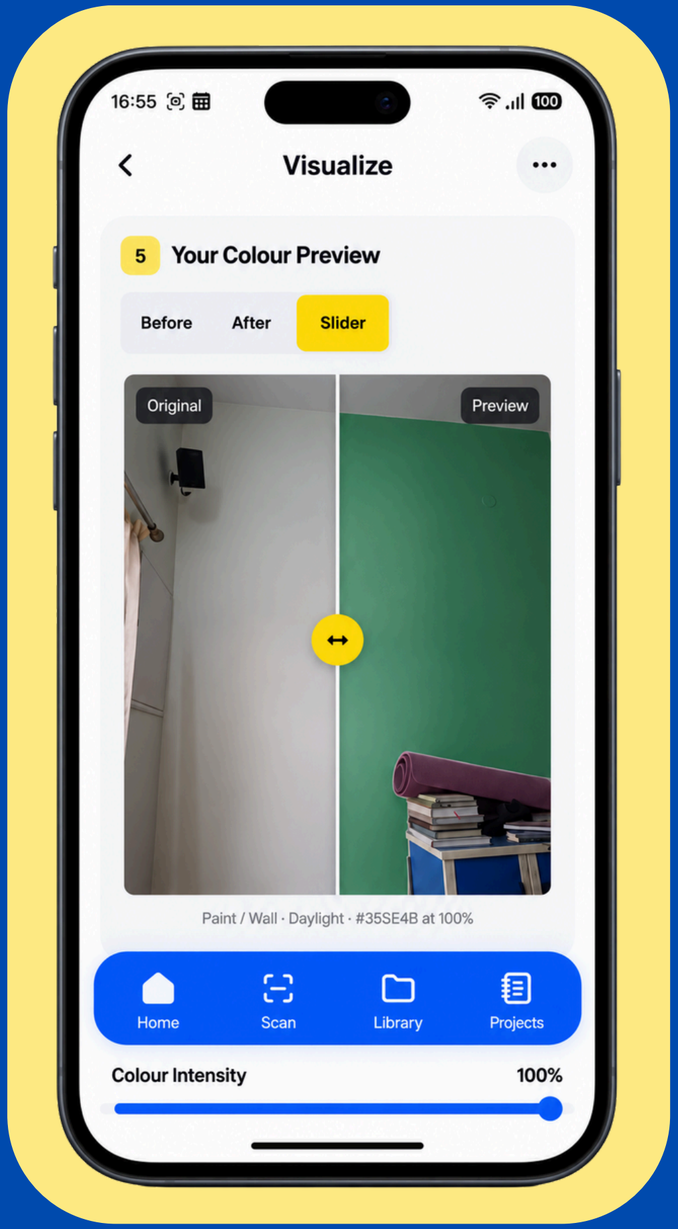
THE FEATURES

- SCAN
- COMPARE
- CONVERT
- IMAGE MATCH
- LIBRARY
- PROJECTS
- VISUALIZE

VISUALIZE COLOUR

HOW IT WORKS

- Upload or capture a photo
- Select the desired surface
- Apply a scanned or saved colour
- Preview the recoloured result



WHO IS CHROMA FOR?

GRAPHICS & DIGITAL ART

Create palettes from real-world colours.

FASHION & TEXTILES

Identify and compare fabric shades.

INTERIORS & DÉCOR

Match and preview wall colours.

DESIGN STUDENTS

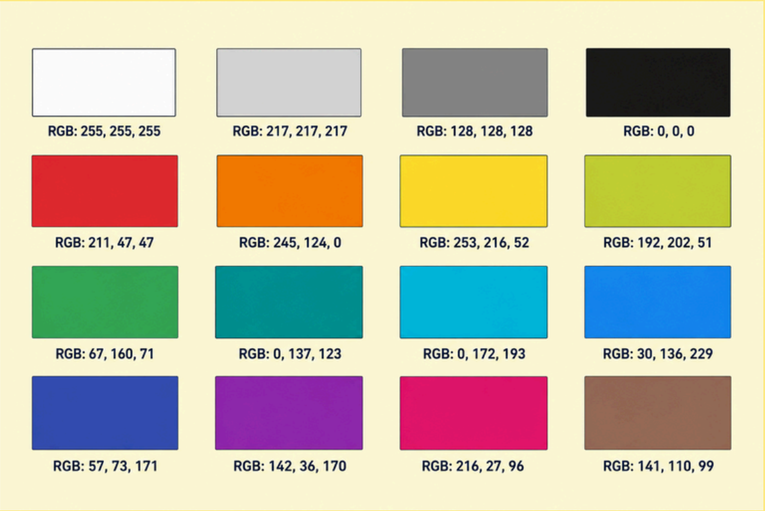
Explore colour matching and conversion.

PRINT & COLOUR

Capture and save colour references.

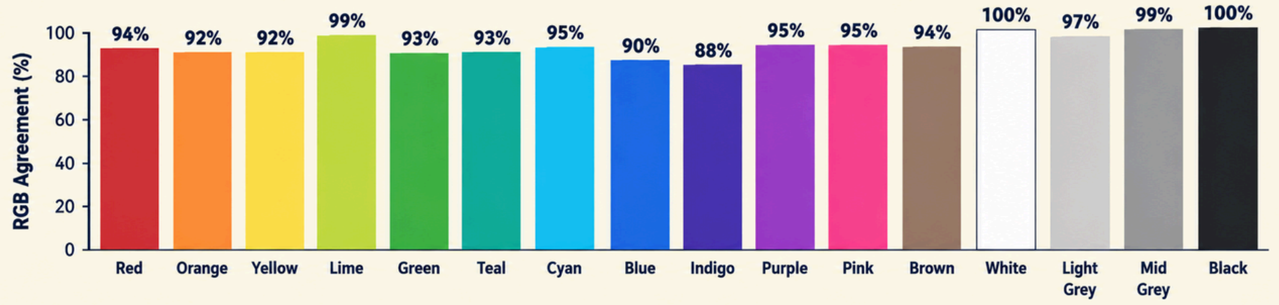
PERFORMANCE ANALYSIS

CALIBRATION SETUP



- 16 RGB reference colours used for calibration
- 5 repeated scans recorded for each colour
- Sensor output compared with intended RGB values

COLOUR DETECTION PERFORMANCE



KEY FINDINGS

94.68%
Average Accuracy

16 RGB
Colours Tested

Consistent Readings

FURTHER IMPROVEMENT
More colours & surfaces for calibration

RGB agreement is measured against intended digital values of printed A4 swatches, not certified colour accuracy.

WHATS NEXT?

COMPACT HARDWARE

Custom PCB and 3D-printed enclosure for a smaller, portable design.

GUIDED CALIBRATION

Simplified calibration process for improved usability.

SOFTWARE INTEGRATIONS

Plugins for tools such as Figma, Photoshop and Canva.

ENHANCED ACCURACY

Expanded calibration and testing across more colours and surfaces.


THANK YOU